Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

The value of $s_n = \sum_{k=1}^{n} \frac{(2)^k(5)^k}{(5^k-2^k)(5^{k+1}-2^{k+1})}$ If $\lim_{n\to\infty}^{Sn}$ is equals

(A) $\frac{1}{9}$

(B) $\frac{2}{9}$

(C) $\frac{3}{9}$

(D) $\frac{5}{9}$

Q.2

If $a_1, a_2, \ldots, a_n$ are in A.P. with common difference d, then the sum of the series $\sin d\left[\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \ldots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n\right]$ is

(A) $\sec a_1 - \sec a_n$

(B) $\operatorname{cosec} a_1 - \operatorname{cosec} a_n$

(C) $\cot \mathrm{a}_1 - \cot \mathrm{a}_\mathrm{n}$

(D) $\tan \mathrm{a}_1 - \tan \mathrm{a}_\mathrm{n}$

Q.3

If a, b, c, d are in G.P., then $\left(a^3+b^3\right)^{-1}, \left(b^3+c^3\right)^{-1}, \left(c^3+d^3\right)^{-1}$ are in-

(A) A.P.

(B) G.P.

(C) H.P.

(D) None of these

Q.4

$\sum_{r=0}^{\infty} \frac{r-1}{2^{r+1}}$ is equal to-

(A) $\frac{1}{2}$

(B) 0

(C) 2

(D) $-\frac{1}{2}$

@IITJEE_Advanced

Q.5

If x, y, z are in A.P. and x, y, t are in G.P. then $x, x-y, t-z$ are in

(A) G.P.

(B) A.P.

(C) H.P.

(D) A.P. and G.P. both

Q.6

If $\mathrm{d}, \mathrm{e}, \mathrm{f}$ are in G.P. and two quadratic equations $\mathrm{ax}^2+2\mathrm{bx}+\mathrm{c}=0$ and $\mathrm{dx}^2+2\mathrm{ex}+\mathrm{f}=0$ have a common root then, $\frac{\mathrm{d}}{\mathrm{a}}, \frac{\mathrm{e}}{\mathrm{b}}, \frac{\mathrm{f}}{\mathrm{c}}$ are in-

(A) H.P.

(B) G.P.

(C) A.P.

(D) None of these

Q.7

If a, b, c in A.P. and $x=\sum_{n=0}^{\infty} a^n, y=\sum_{n=0}^{\infty} b^n, z=\sum_{n=0}^{\infty} c^n$ then x, y, z are in: [Given $(|\mathrm{a}|<1,|\mathrm{b}|<1,|\mathrm{c}|<1)$]

(A) AP

(B) GP

(C) HP

(D) None of these

Q.8

Let r^{th} term of a series be given by $t_r=\frac{r}{1-3r^2+r^4}$. Then $\lim_{n\to\infty}\sum_{r=1}^{n} t_r$ is

(A) $3/2$

(B) $1/2$

(C) $-1/2$

(D) $-3/2$

Q.9

If $x^2+9y^2+25z^2=xyz\left(\frac{15}{x}+\frac{5}{y}+\frac{3}{z}\right)$, then $-$

(A) x, y and z are in H.P.

(B) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

@IITJEE_Advanced

(C) x, y, z are in G.P.

(D) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in G.P.

Q.10

Given that $x + y + z = 15$ when a, x, y, z, b are in A.P. and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$ when a, x, y, z, b are in H.P. Then -

(A) G.M. of a and b is 3

(B) One possible value of $a + 2b$ is 11

(C) A.M. of a and b is 6

(D) Greatest value of $a - b$ is 8

Q.11

The terms of an infinitely decreasing G.P. in which all the terms are positive, the first term is 4, and the difference between the third and fifth term is $32/81$, then -

(A) $r = 1/3$

(B) $r = 2\sqrt{2}/3$

(C) $S_\infty = 6$

(D) None of these

Q.12

Passage:
An arithmetic progression (A.P.) is a sequence in which terms increase or decrease regularly by the same constant. A sequence is said to be a geometric progression or G.P., if the ratio of any term to its preceding term is same throughout. Harmonical progression is defined as a series in which reciprocal of its terms are in A.P.
Question:
If the roots of the equation $\left(a^2 + b^2\right) x^2 - 2b(a + c)x + \left(b^2 + c^2\right) = 0$ are real and equal, then

(A) a+b+c = 0

(B) a, b, c are in A.P.

(C) a, b, c are in G.P.

(D) a, b, c are in H.P.

Q.13

Passage:
An arithmetic progression (A.P.) is a sequence in which terms increase or decrease regularly by the same constant. A sequence is said to be a geometric progression or G.P., if the ratio of any term to its preceding term is same throughout. Harmonical progression is defined as a series in which reciprocal of its terms are in A.P.

@IITJEE_Advanced

Question:

If the equation $ax^2 + bx + c = 0$ and $x^3 - 3x^2 + 3x - 1 = 0$ have a common real root, then $-$

(A) $a + b + c = 0$

(B) a, b, c are in A.P.

(C) a, b, c are in G.P.

(D) a, b, c are in H.P.

Q.14

Passage:

An arithmetic progression (A.P.) is a sequence in which terms increase or decrease regularly by the same constant. A sequence is said to be a geometric progression or G.P., if the ratio of any term to its preceding term is same throughout. Harmonical progression is defined as a series in which reciprocal of its terms are in A.P.

Question:

Let a,b,c be positive real numbers such that the expression

$\mathrm{bx}^2 + \sqrt{\left((\mathrm{a}+\mathrm{c})^2 + 4\,\mathrm{b}^2\right)}\mathrm{x} + (\mathrm{a}+\mathrm{c})$ is non-negative $\forall x \in R$, then-

(A) $a + b + c = 0$

(B) a, b, c are in A.P.

(C) a, b, c are in G.P.

(D) a, b, c are in H.P.

Q.15

Passage:

In a G.P., the sum of the first and last term is 66, the product of the second and the second last one is 128 and the sum of the terms is 126 .

Question:

If an increasing G.P. is considered, then the number of terms in G.P. is $-$

(A) 9

(B) 8

(C) 12

(D) 6

Q.16

@IITJEE_Advanced

Passage:

In a G.P., the sum of the first and last term is 66, the product of the second and the second last one is 128 and the sum of the terms is 126 .

Question:

If the decreasing G.P. is considered, then the sum of infinite terms is -

(A) 64

(B) 128

(C) 256

(D) 729

Q.17

Passage:

In a G.P., the sum of the first and last term is 66, the product of the second and the second last one is 128 and the sum of the terms is 126 .

Question:

In any case, the difference between least and greatest term is

(A) 78

(B) 126

(C) 127

(D) None of these

Q.18

@IITJEE_Advanced

Match the column -

Column I	Column II
(A) If $\Sigma n = 210$, then Σn^2 is divisible by the greatest prime number which is greater than	(p) 16
(B) Between 4 and 2916 $(2n+1)$ GM.'s are inserted. Then the $(n+1)$th G.M. is divisible by greatest odd integer which is less than	(q) 10
(C) In a certain progression, three consecutive terms are 40, 30, 24, 20. Then the integral part of the next term of the progression is more than	(r) 34
(D) $1+\frac{4}{5}+\frac{7}{5^2}+\frac{10}{5^3}+.....\text{to}\ \infty=\frac{a}{b}$, where H.C.F. (a, b) = 1, then a – b is less than	(s) 30

(A) (A) - p,r,s, (B) - r,s, (C) - p,q, (D) - q,r

(B) (A) - q, (B) - p,q,r,s, (C) - p,q, (D) - q,r

(C) (A) - p,r,s, (B) - q,r, (C) - p,r,s, (D) - r,s

(D) (A) - p,q,r,s, (B) - r,s, (C) - p,q, (D) - r,s

Q.19

If pth, qth and rth terms of a H.P. be respectively a, b and c then find the value of (q - r)bc + (r - p) ca + (p - q) ab

Q.20

If $\sum_{i=1}^{503} x_i^2 = 1006$
then minimum value of $\sum_{i=1}^{503} \frac{8}{503 \cdot x_i^2}$ is

Answers & Solutions

Q.1 Answer:

$\frac{2}{9}$

Solution:

$$S_n = \sum_{k=1}^{n} \frac{\left(\frac{2}{5}\right)^k}{\left(1-\left(\frac{2}{5}\right)^k\right)\left(5-2\left(\frac{2}{5}\right)^k\right)} = \sum_{k=1}^{n} \left(\frac{p}{(1-p)(5-2p)}\right)$$

$$= \sum_{k=1}^{n} \left(\frac{1}{3(1-p)} - \frac{5}{3(5-2p)}\right) = \sum_{k=1}^{n} \left(\frac{1}{3(1-p)} - \frac{1}{3\left(1-\frac{2}{5}p\right)}\right)$$

$$= \frac{1}{3}\sum_{k=1}^{n} \left(\frac{1}{\left(1-\left(\frac{2}{5}\right)^k\right)} - \frac{1}{\left(1-\left(\frac{2}{5}\right)^{k+1}\right)}\right) = \frac{1}{3}\sum_{k=1}^{n} (v_k - v_{k+1})$$

@IITJEE_Advanced

$= \frac{1}{3}\left[v_k - v_{k+1}\right) = \frac{1}{3}\left[\frac{5}{3} - \frac{1}{1-\left(\frac{2}{5}\right)^{k+1}}\right]$

$\lim_{x\to\infty} S_n = \frac{1}{3}\left[\frac{5}{3} - 1\right] = \frac{2}{9}$

Q.2 Answer:

$\cot a_1 - \cot a_n$

Solution:

We have $d = a_2 - a_1 = a_3 - a_2 = \ldots = a_n - a_{n-1}$.

Thus, $\sin d\left[\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \ldots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n\right]$

$= \frac{\sin(a_2-a_1)}{\sin a_1 \sin a_2} + \frac{\sin(a_3-a_2)}{\sin a_2 \sin a_3} + \ldots + \frac{\sin(a_n-a_{n-1})}{\sin a_{n-1} \sin a_n}$

$= (\cot a_1 - \cot a_2) + (\cot a_2 - \cot a_3) + \ldots + (\cot a_{n-1} - \cot a_n) = \cot a_1 - \cot a_n$

Q.3 Answer:

G.P.

Solution:

Let $b = ar, c = ar^2$ **and** $d = ar^3$**. Then,**

$\frac{1}{a^3+b^3} = \frac{1}{a^3(1+r^3)}, \frac{1}{b^3+c^3} = \frac{1}{a^3r^3(1+r^3)}$ **and,** $\frac{1}{c^3+d^3} = \frac{1}{a^3r^6(1+r^3)}$

Clearly, $\left(a^3 + b^3\right)^{-1}, \left(b^3 + c^3\right)^{-1}$ **and** $\left(c^3 + d^3\right)^{-1}$ **are in G.P. with common ratio** $1/r^3$**.**

Q.4 Answer:

0

Solution:

$S = -\frac{1}{2} + \frac{0}{2^2} + \frac{1}{2^3} + \frac{2}{2^4} + \ldots$ **upto infinity i.e.**

A.G.P.

$\frac{S}{2} = -\frac{1}{2^2} + \frac{0}{2^3} + \frac{1}{2^4} + \ldots\ldots$

$\frac{S}{2} = -\frac{1}{2} + \left[\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots\right.$ **upto infinity** $\left.\right]$

$= -\frac{1}{2} + \frac{1/4}{1-\frac{1}{2}} \Rightarrow S = 0$

Q.5 Answer:

G.P.

Solution:

x, y, z **are in A.P.**

$\Rightarrow \quad 2y = x + z$

or $\quad 2xy = x^2 + xz$

(multiplying with x**)**

$\Rightarrow \quad x^2 - 2xy = -xz$ **....(1)**

x, y, t **are in G.P.**

@IITJEE_Advanced

$\Rightarrow \quad y^2 = xt \ldots (2)$

or $\quad \left(x^2 - 2xy + y^2\right) = -xz + xt$

or $\quad (x-y)^2 = x(t-z)$

$\mathrm{x}, \mathrm{x}-\mathrm{y}, \mathrm{t}-\mathrm{z}$ **are in G.P.**

Q.6 Answer:

H.P.

Solution:

Here $\mathrm{e}^2 = \mathrm{df}$

Now $dx^2 + 2ex + f = 0$ **given**

$\Rightarrow \quad \mathrm{dx}^2 + 2\sqrt{\mathrm{df}}\mathrm{x} + \mathrm{f} = 0 \Rightarrow \mathrm{x} = -\sqrt{\frac{\mathrm{f}}{\mathrm{d}}}$

Putting in $\mathrm{ax}^2 + 2bx + c = 0$ **we get**

$\mathrm{a}\frac{\mathrm{f}}{\mathrm{d}} + \mathrm{c} = 2\,\mathrm{b}\sqrt{\frac{\mathrm{f}}{\mathrm{d}}} \Rightarrow \frac{\mathrm{a}}{\mathrm{d}} + \frac{\mathrm{c}}{\mathrm{f}} = \frac{2\,\mathrm{b}}{\mathrm{e}}$

$\therefore \frac{\mathrm{a}}{\mathrm{d}}, \frac{\mathrm{b}}{\mathrm{e}}, \frac{\mathrm{c}}{\mathrm{f}}$ **are in A.P.;**

$\frac{\mathrm{d}}{\mathrm{a}}, \frac{\mathrm{e}}{\mathrm{b}}, \frac{\mathrm{f}}{\mathrm{c}}$ **are in H.P.**

Q.7 Answer:

HP

Solution:

Here a, b, c **in A.P., given**

Also $x = \frac{1}{1-a}, y = \frac{1}{1-b}, z = \frac{1}{1-c}$

Now a,b,c in AP

$\Rightarrow 1-\mathrm{a}, 1-\mathrm{b}, 1-\mathrm{c}$ **in A.P.**

$\Rightarrow \quad \frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ **in H.P.**

$\Rightarrow x, y, z$ **in H.P**

Q.8 Answer:

$-1/2$

Solution:

$\sum_{r=1}^{n} t_r = \sum_{r=1}^{n} \frac{r}{1-3r^2+r^4} = \sum_{r=1}^{n} \frac{r}{\left[(r^2-1)^2 - r^2\right]}$

$$\sum_{r=1}^{n} \frac{1}{2} \frac{\left(r^2+r-1\right) - \left(r^2-r-1\right)}{\left(r^2-r-1\right)\left(r^2+r-1\right)}$$

$$= \frac{1}{2} \sum_{r=1}^{n} \left[\frac{1}{r^2-r-1} - \frac{1}{r^2+r-1}\right] = \frac{1}{2}\left[-1 - \frac{1}{\left(n^2+n-1\right)}\right]$$

$$= \frac{1}{2}\left[\frac{-n^2-n+1-1}{n^2+n-1}\right] = \frac{1}{2}\left[\frac{-n^2-n}{n^2+n-1}\right]; \lim_{n\to\infty} \sum_{r=1}^{n} t_r = \frac{-1}{2}$$

Q.9 Answer:

x, y and z are in H.P.

@IITJEE_Advanced

Solution:

$$x^2+9y^2+25z^2=15yz+5zx+3xy$$
$$\Rightarrow (x)^2+(3y)^2+(5z)^2-(x)(3y)-(3y)(5z)-(x)(5z)=0$$
$$\Rightarrow \frac{1}{2}\left[(x-3y)^2+(3y-5z)^2+(x-5z)^2\right]=0$$
$$\Rightarrow x-3y=0, 3y-5z=0, x-5z=0$$
$$\Rightarrow \frac{1}{x}=\frac{1}{3y} \text{ and } \frac{5}{3y}=\frac{1}{z} \text{ and } \frac{1}{5z}=\frac{1}{x}$$
$$\Rightarrow \frac{1}{x}+\frac{1}{z}=\frac{1}{3y}+\frac{5}{3y}=\frac{2}{y}$$

Therefore, $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P. and x, y and z are in H.P.

Q.10 Answer:

G.M. of a and b is 3

Solution:

$$x+y+z=3\left(\frac{a+b}{2}\right) \Rightarrow 15=3\left(\frac{a+b}{2}\right)$$
$$\Rightarrow a+b=10....(1)$$
$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{3\left(\frac{1}{a}+\frac{1}{b}\right)}{2} \Rightarrow \frac{5}{3}=\frac{3(a+b)}{2ab}=\frac{3\times 10}{2ab}$$
$$\Rightarrow \quad ab=9 \text{ ...(2)}$$

From eq. (1) and $(2), a=9, b=1$ or $a=1$ and $b=9$

Hence, G.M. $=\sqrt{ab}=3, a+2b=11$ or 19

Q.11 Answer:

$r=1/3$

Solution:

Given that $a=4, T_3-T_5=32/81$.

Hence, $a\left(r^2-r^4\right)=32/81$ or $\quad r^4-r^2+8/81=0$ or $81r^4-81r^2+8=0$ or

$$\left(9r^2-8\right)\left(9r^2-1\right)=0$$
$$\therefore \quad r^2=8/9, 1/9$$

Therefore, the value of r is to be +ve since all the terms are +ve.

For $r=1/3$,

$$S_\infty=\frac{a}{1-r}=\frac{4}{1-\frac{1}{3}}=\frac{4\times 3}{2}=6$$

Similarly, we can find S_∞ when $r=2\sqrt{2}/3$

Q.12 Answer:

a, b, c are in G.P.

Solution:

(i) $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$

The roots are real and equal.

@IITJEE_Advanced

Hence, $4b^2(a+c)^2 - 4\left(a^2+b^2\right)\left(b^2+c^2\right) = 0$

$\Rightarrow \quad b^2\left(a^2+c^2+2ac\right) - \left(a^2b^2+a^2c^2+b^4+b^2c^2\right) = 0$

$\Rightarrow \quad b^2a^2+b^2c^2+2ab^2c-a^2b^2-a^2c^2-b^4-b^2c^2 = 0$

$\Rightarrow \quad 2ab^2c - a^2c^2 - b^4 = 0 \Rightarrow \left(b^2-ac\right)^2 = 0$

Hence, $b^2 = ac$**. Thus,** a, b, c **are in G.P.**

(ii) $(x-1)^3 = 0 \Rightarrow x = 1$ **is the common root. Hence,** $a+b+c = 0$.

(iii) $(a+c)^2 + 4b^2 - 4b(a+c) \leq 0, x \in R$

$\Rightarrow \quad ((a+c)-2b)^2 \leq 0$

$\Rightarrow a+c = 2\,b \Rightarrow a, b, c$ **in A.P.**

Q.13 Answer:

$a+b+c = 0$

Solution:

(i) $\left(a^2+b^2\right)x^2 - 2b(a+c)x + \left(b^2+c^2\right) = 0$

The roots are real and equal.

Hence, $4b^2(a+c)^2 - 4\left(a^2+b^2\right)\left(b^2+c^2\right) = 0$

$\Rightarrow \quad b^2\left(a^2+c^2+2ac\right) - \left(a^2b^2+a^2c^2+b^4+b^2c^2\right) = 0$

$\Rightarrow \quad b^2a^2+b^2c^2+2ab^2c-a^2b^2-a^2c^2-b^4-b^2c^2 = 0$

$\Rightarrow \quad 2ab^2c - a^2c^2 - b^4 = 0 \Rightarrow \left(b^2-ac\right)^2 = 0$

Hence, $b^2 = ac$**.**

Thus, a, b, c **are in G.P.**

(ii) $(x-1)^3 = 0 \Rightarrow x = 1$ **is the common root. Hence,** $a+b+c = 0$.

(iii) $(a+c)^2 + 4b^2 - 4b(a+c) \leq 0, x \in R$

$\Rightarrow \quad ((a+c)-2b)^2 \leq 0$

$\Rightarrow a+c = 2\,b \Rightarrow a, b, c$ **in A.P.**

Q.14 Answer:

a, b, c are in A.P.

Solution:

(i) $\left(a^2+b^2\right)x^2 - 2b(a+c)x + \left(b^2+c^2\right) = 0$

The roots are real and equal.

Hence, $4b^2(a+c)^2 - 4\left(a^2+b^2\right)\left(b^2+c^2\right) = 0$

$\Rightarrow \quad b^2\left(a^2+c^2+2ac\right) - \left(a^2b^2+a^2c^2+b^4+b^2c^2\right) = 0$

$\Rightarrow \quad b^2a^2+b^2c^2+2ab^2c-a^2b^2-a^2c^2-b^4-b^2c^2 = 0$

$\Rightarrow \quad 2ab^2c - a^2c^2 - b^4 = 0 \Rightarrow \left(b^2-ac\right)^2 = 0$

Hence, $b^2 = ac$**.**

Thus, a, b, c **are in G.P.**

(ii) $(x-1)^3 = 0 \Rightarrow x = 1$ **is the common root. Hence,** $a+b+c = 0$.

(iii) $(a+c)^2 + 4b^2 - 4b(a+c) \leq 0, x \in R$

$\Rightarrow \quad ((a+c)-2b)^2 \leq 0$

$\Rightarrow a+c = 2\,b \Rightarrow a, b, c$ **in A.P.**

@IITJEE_Advanced

Q.15 Answer:

6

Solution:

Let a be the first term and r the common ratio of the given G.P. Further, let there be n terms in the given G.P. Then,

$$\begin{aligned} &a_1 + a_n = 66 \Rightarrow a + ar^{n-1} = 66 \ldots (1) \\ &a_2 \times a_{n-1} = 128 \end{aligned}$$

$a_2 \times a_{n-1} = 128$

$\Rightarrow ar \times ar^{n-2} = 128 \Rightarrow a^2 r^{n-1} = 128 \Rightarrow a \times \left(ar^{n-2}\right) = 128 \Rightarrow ar^{n-1} = \frac{128}{a}$

Putting this value of ar^{n-1} in (1), we get

$$a + \frac{128}{a} = 66 \Rightarrow a^2 - 66a + 128 = 0 \Rightarrow (a-2)(a-64) = 0 \Rightarrow a = 2, 64$$

Putting $a = 2$ in (1), we get $2 + 2 \times r^{n-1} = 66 \Rightarrow r^{n-1} = 32$

Putting a $= 64$ in (1), we get $64 + 64r^{n-1} = 66 \Rightarrow r^{n-1} = \frac{1}{32}$

For an increasing G.P., $r > 1$. Now, $S_n = 126$

$2\left(\frac{r^n - 1}{r-1}\right) = 126 \Rightarrow \frac{r^n - 1}{r-1} = 63 \Rightarrow \frac{r^{n-1} \times r - 1}{r-1} = 63 \Rightarrow \frac{32r-1}{r-1} = 63 \Rightarrow r = 2$

@IITJEE_Advanced

$\therefore \quad r^{n-1} = 32 \Rightarrow 2^{n-1} = 32 = 2^5 \Rightarrow n - 1 = 5 \Rightarrow n = 6$

For decreasing G.P., $a = 64$ and $r = 1/2$

Hence, the sum of infinite terms is $64/\{1 - (1/2)\} = 128$

Q.16 Answer:

128

Solution:

Let a be the first term and r the common ratio of the given G.P. Further, let there be n terms in the given G.P. Then,

$$\begin{aligned} &a_1 + a_n = 66 \Rightarrow a + ar^{n-1} = 66 \ldots (1) \\ &a_2 \times a_{n-1} = 128 \end{aligned}$$

$a_2 \times a_{n-1} = 128$

$\Rightarrow ar \times ar^{n-2} = 128 \Rightarrow a^2 r^{n-1} = 128 \Rightarrow a \times \left(ar^{n-2}\right) = 128 \Rightarrow ar^{n-1} = \frac{128}{a}$

Putting this value of ar^{n-1} in (1), we get

$$a + \frac{128}{a} = 66 \Rightarrow a^2 - 66a + 128 = 0 \Rightarrow (a-2)(a-64) = 0 \Rightarrow a = 2, 64$$

@IITJEE_Advanced

Putting $a = 2$ in (1), we get $2 + 2 \times r^{n-1} = 66 \Rightarrow r^{n-1} = 32$

Putting a $= 64$ in (1), we get $64 + 64r^{n-1} = 66 \Rightarrow r^{n-1} = \frac{1}{32}$

For an increasing G.P., $r > 1$. Now, $S_n = 126$

$2\left(\frac{r^n-1}{r-1}\right) = 126 \Rightarrow \frac{r^n-1}{r-1} = 63 \Rightarrow \frac{r^{n-1}\times r-1}{r-1} = 63 \Rightarrow \frac{32r-1}{r-1} = 63 \Rightarrow r = 2$

$\therefore \quad r^{n-1} = 32 \Rightarrow 2^{n-1} = 32 = 2^5 \Rightarrow n - 1 = 5 \Rightarrow n = 6$

For decreasing G.P., $a = 64$ and $r = 1/2$

Hence, the sum of infinite terms is $64/\{1 - (1/2)\} = 128$

Q.17 Answer:

None of these

Solution:

Let a be the first term and r the common ratio of the given G.P. Further, let there be n terms in the given G.P. Then,

$$\begin{aligned} &a_1 + a_n = 66 \Rightarrow a + ar^{n-1} = 66 \ldots (1) \\ &a_2 \times a_{n-1} = 128 \end{aligned}$$

$a_2 \times a_{n-1} = 128$

@IITJEE_Advanced

$\Rightarrow ar \times ar^{n-2} = 128 \Rightarrow a^2 r^{n-1} = 128 \Rightarrow a \times (ar^n) = 128 \Rightarrow ar^{n-1} = \frac{128}{a}$

Putting this value of ar^{n-1} in (1), we get

$$a + \frac{128}{a} = 66 \Rightarrow a^2 - 66a + 128 = 0 \Rightarrow (a-2)(a-64) = 0 \Rightarrow a = 2, 64$$

Putting $a = 2$ in (1), we get $2 + 2 \times r^{n-1} = 66 \Rightarrow r^{n-1} = 32$

Putting a $= 64$ in (1), we get $64 + 64r^{n-1} = 66 \Rightarrow r^{n-1} = \frac{1}{32}$

For an increasing G.P., $r > 1$. Now, $S_n = 126$

$2\left(\frac{r^n - 1}{r-1}\right) = 126 \Rightarrow \frac{r^n - 1}{r-1} = 63 \Rightarrow \frac{r^{n-1} \times r - 1}{r-1} = 63 \Rightarrow \frac{32r-1}{r-1} = 63 \Rightarrow r = 2$

$\therefore \quad r^{n-1} = 32 \Rightarrow 2^{n-1} = 32 = 2^5 \Rightarrow n - 1 = 5 \Rightarrow n = 6$

For decreasing G.P., $a = 64$ and $r = 1/2$

Hence, the sum of infinite terms is $64/\{1 - (1/2)\} = 128$

Q.18 Answer:

(A) - p,q,r,s, (B) - r,s, (C) - p,q, (D) - r,s

Solution:

@IITJEE_Advanced

(A) $-pqrs, (B) - rs, (C) - pq, (D) - rs$

(A) $\Sigma n = 210 \Rightarrow n(n+1) = 420$

$\Rightarrow (n-20)(n+21) = 0 \Rightarrow n = 20$

Hence, $\Sigma n^2 = \frac{n}{6}(n+1)(2n+1) = \frac{20}{6}(21)(41) = (10)(7)(41)$

Hence, the greatest prime number by which Σn^2 **is divisible is 41.**

(B) $4, G_1, G_2, \ldots\ldots G_{n+1}, \ldots\ldots G_{2n}, G_{2n+1}, 2916$

G_{n+1} **will be the middle mean of** $(2n+1)$ **odd means and it will be equidistance from the first and last terms. Hence,**

$4, G_{n+1}, 2916$ will also be in G.P. So,

$\Rightarrow {G^2}_{n+1} = 4 \times 2916 = 4 \times 9 \times 324 = 4 \times 9 \times 4 \times 81$

$\Rightarrow G_{n+1} = 2 \times 3 \times 2 \times 9 = 108$

Hence, the greatest odd number by which G_{n+1} **is divisible is 27.**

(C) Terms are 40, 30, 24, 20.

Now, $\frac{1}{30} - \frac{1}{40} = \frac{1}{120}$

$\frac{1}{24} - \frac{1}{30} = \frac{6}{24 \times 30} = \frac{1}{120}$ **and** $\frac{1}{20} - \frac{1}{24} = \frac{4}{20 \times 24} = \frac{1}{120}$

Hence, $1/30, 1/24, 1/20$ **are in A.P. with common difference** $d = 1/120$.

Hence, the next term is $\frac{1}{20} + \frac{1}{120} = \frac{7}{120}$.

Therefore, the next term of given series is $\frac{120}{7} = 17\frac{1}{7}$.

Hence, the integral part of $17\frac{1}{7}$ **is 17.**

(D) $S = 1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots\ldots\infty \Rightarrow \frac{1}{5}S = \frac{1}{5} + \frac{4}{5^2} + \frac{7}{5^3} + \ldots\ldots$

$\Rightarrow S\left(1 - \frac{1}{5}\right) = 1 + 3\left[\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \ldots\infty\right] \Rightarrow \frac{4}{5}S = 1 + 3\left[\frac{1/5}{1-1/5}\right] = 1 + \frac{3}{4} = \frac{7}{4}$

$\Rightarrow S = \frac{35}{16}$

$\Rightarrow a = 35$ **and** $b = 16 \Rightarrow a - b = 19$

Q.19 Answer:

—

Solution:

Let A and D be the first term and common difference of the corresponding A.P.

Now a, b, c are respectively the pth, qth and rth terms of H.P.

$\therefore$ $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ **will be respectively the pth, qth and rth terms of the corresponding A.P.**

$\Rightarrow \quad \frac{1}{a} = A + (p-1)D$

$\frac{1}{b} = A + (q-1)D$

$\frac{1}{c} = A + (r-1)D$

Subtracting (3) from (2) we get $\frac{1}{b} - \frac{1}{c} = (q-r)D$

$\Rightarrow \quad bc(q-r) = \frac{c-b}{D} = -\frac{(b-c)}{D}$

@IITJEE_Advanced

$\therefore \quad \text{L.H.S.} = \Sigma bc(q - r) = -\frac{1}{D}\Sigma(b - c)$

$= -\frac{1}{D}[b - c + c - a + a - b] = 0 = \text{R.H.S.}$

Q.20 Answer:

—

Solution:

$$\text{A.M} \geq \text{H.M}$$

$$\frac{x_1^2 + x_2^2 + \ldots\ldots + x_{503}^2}{503} \geq \frac{503}{\frac{1}{x_1^2} + \frac{1}{x_2^2} + \ldots\ldots + \frac{1}{x_{503}^2}}$$

$$\Rightarrow \frac{\sum x_i^2}{503} \geq \frac{503}{\sum_{i=1}^{503} \frac{1}{x_i^2}} \qquad \Rightarrow \sum_{i=1}^{503} \frac{1}{x_i^2} \geq \frac{503}{2}$$

$\Rightarrow \sum_{i=1}^{503} \frac{8}{503 \cdot x_i^2} \geq 4$

@IITJEE_Advanced